

Nicole Thiessen

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EDUCATION

University of Waterloo

Bachelor of Applied Science in Mechatronics Engineering

Waterloo, ON

Expected 2030

TECHNICAL SKILLS

Electrical: Battery Systems, High-Voltage Safety, Soldering, Wire Harness Design and Assembly, HIL Bench Testing

Mechanical: SolidWorks, AutoCAD, GD&T, CFD Simulations, DFM

Software: C++, Python, CANalyzer, Git, TensorFlow, PyTorch, OpenCV

EXPERIENCE

Vehicle Systems Engineering Intern

May 2026 – August 2026

Geotab

Waterloo, ON

- Decoded proprietary vehicle CAN bus signals using CANalyzer and scan tools, expanding Geotab's telematics support
- Designed and ran hardware-in-the-loop bench simulations, wiring physical CAN networks to emulate vehicle-device interaction and validate firmware before fleet-wide deployment
- Root-caused telematics data failures by developing custom diagnostic queries and analyzing remote fleet data, ensuring signal quality across millions of devices

Technical Analysis and Intellectual Property Assessment Intern

June 2025 – August 2025

CoreCurrent Solutions Inc.

Ottawa, ON

- Reverse engineered automotive electronic modules such as ADAS Controls and Front Radar Sensors to identify patent infringements, accelerating legal evidence collection
- De-soldered and researched IC components, compiling findings into organized reports, improving accuracy in client cases
- Built a Python/OpenCV tool to automatically map chip images to PCB positions, streamlining IC-identification workflows for the team

PROJECTS

Formula Electric Design Team - Battery Team Lead

Sept 2025 - Present

- Responsible for the redesign of next year's battery electronics architecture, optimizing the high-voltage wire harness and low-voltage safety systems to improve pack reliability
- Developed a cell-grouping algorithm for battery segments, using cell DCIR data to balance power output and long-term battery health
- Manufactured lithium-ion battery segments through epoxy bonding, resistance spot welding, and AMS board assembly

AI Voice Assistant

August 2025

- Built an AI voice assistant using Whisper and an LLM, enabling real-time conversations
- Modeled and rigged a 3D robot avatar that reacted to emotions. Designed a custom neural network using PyTorch to map emotion metrics to joint angles, delivering human-like AI interaction

Synthetic Data Generator

July 2025

- Built a synthetic data generator combining CTGAN and LLM enrichment into a pipeline, producing large (10k+ sample) flexible datasets to address AI training data scarcity

UW Crowd – Hack the North

Sept 2025

- Developed a real-time campus occupancy website using Raspberry Pis to scan Bluetooth Low Energy signals, helping students plan around crowds

AWARDS & ACHIEVEMENTS

Academic: 95.5% , 43/45 score on the IB Diploma

Governor General Academic Bronze Medal: Highest academic average, 250+ students

Colonel By S.S. Graduate Award: Academic, athletic, and leadership excellence